

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BASIC MATH AND CALCULUS

LECTURE-3

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Revision
- 2** Trigonometry
- 3** Equation
- 4** Important Question
- 5** ..

Recap From 1st Lec.

Maths

- Maths Bohat Aasaan hai
- number
- + - x ÷
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

$$\frac{1}{0} = \infty$$

$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$\theta = \frac{\text{Arc}}{\text{Radius}}$$

0 Rad.

$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

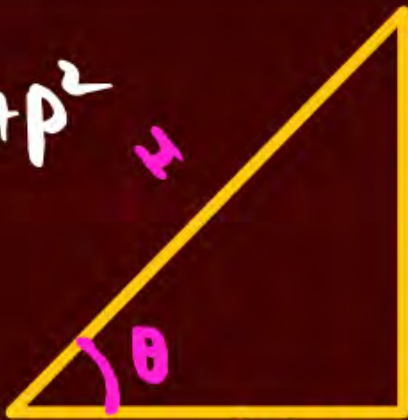
Recap From 2nd Lec.



Trigonometry

Special Δ

• $H^2 = B^2 + P^2$



P/O

B/A

$$\begin{aligned}\sin \theta &= \frac{P}{H} \\ \cos \theta &= \frac{B}{H} \\ \tan \theta &= \frac{P}{B}\end{aligned}$$

$$\begin{aligned}\operatorname{cosec} \theta &= \frac{H}{P} \\ \sec \theta &= \frac{H}{B} \\ \cot \theta &= \frac{B}{P}\end{aligned}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$



P=3

B=4

Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	$\frac{0}{1} = 0$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$\frac{1}{0} = \infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Quadrant System



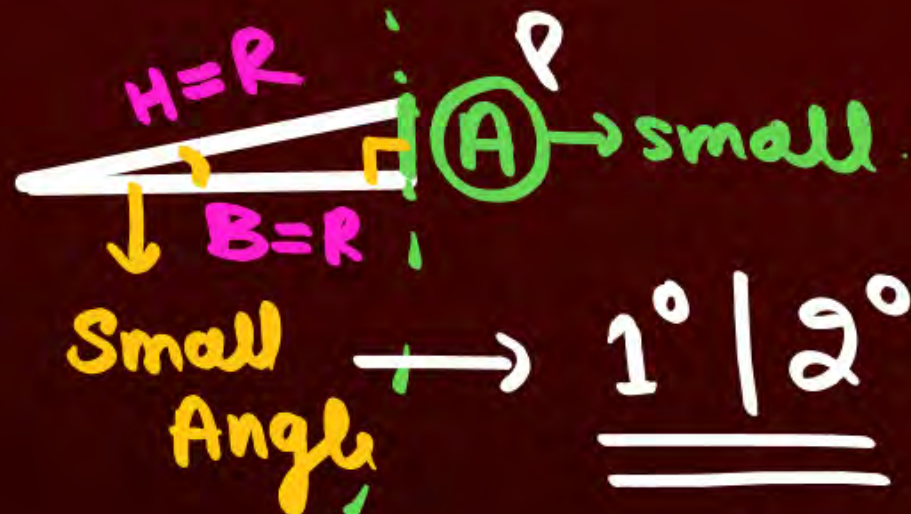
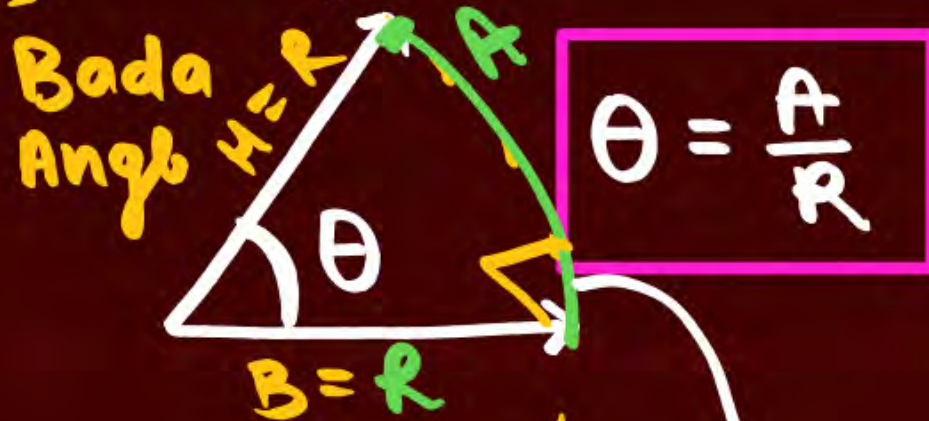
Small Angle Approximation

- Angle → Bohat Bada

$$\# \sin \theta = \frac{P}{H} = \frac{A}{R} = \theta \checkmark$$

$$\# \tan \theta = \frac{P}{B} = \frac{A}{R} = \theta$$

$$\times \# \cos \theta = \frac{B}{H} = \frac{R}{R} = 1$$



1° | 2°

QUESTIONS

Cos 2° is equal

1 ✓

2 2

3 3

4 5

$$\cos 2^\circ$$

$$\cos \theta = 1$$

$$\cos 1^\circ \text{ Small Angle}$$

$$\cos \theta = 1$$

Q ②

Sin 1°

• $\sin \theta = \theta$

$$\sin 1^\circ = 1^\circ = \frac{\pi}{180} \text{ rad}$$

Q ③

Sin 2° = 2°

$$= 2 \times 1^\circ$$

$$= \cancel{2} \times \frac{\pi}{\cancel{180} 90} \text{ rad}$$

$$= \frac{\pi}{90} \text{ rad} \checkmark$$

Equation

- when you write English statement in maths language

(+ - $\times$ $\div$ =)

x, y , numbers

Variable
GF
BF

constt
↓
Parents

Q The cost of Pen^y is twice the cost of pencil^x (make a equation)

Sol Let cost of Pen = y
cost of Pencil = x

ATQ $\text{cop} = 2 \text{cop}$

$$y = 2x$$

Graph
table

Graph

x	1	2	3
y	$y = 2x$ $y = 2 \times 1$ 2	$y = 2x$ $y = 2 \times 2$ $= 4$	$y = 2x$ $= 2 \times 3$ $= 6$

Graph

$$y = 2x$$

Straight line equation

Slope

x	1	2	3	4
y	2	4	6	8

Slope (kitna chhrayi hai kitna utrayi hai)

$$\text{Slope} = \frac{y}{x} = \frac{P}{B} = \tan \theta$$

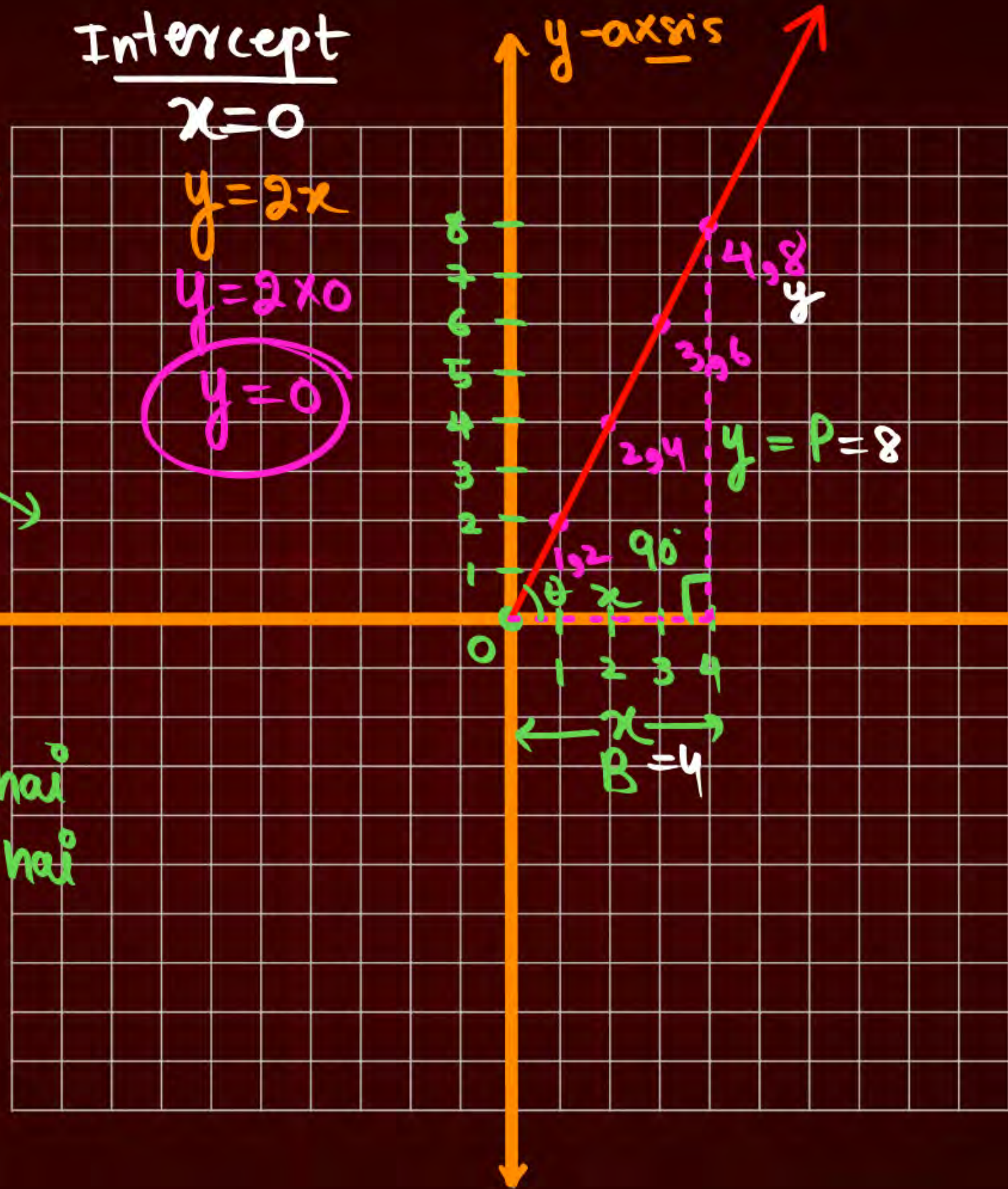
Intercept

$$x=0$$

$$y=2x$$

$$y=2 \times 0$$

$$y=0$$



$$y = 2x$$

Slope = 2

$$\text{Slope} = \frac{y}{x} = \frac{8}{4}$$

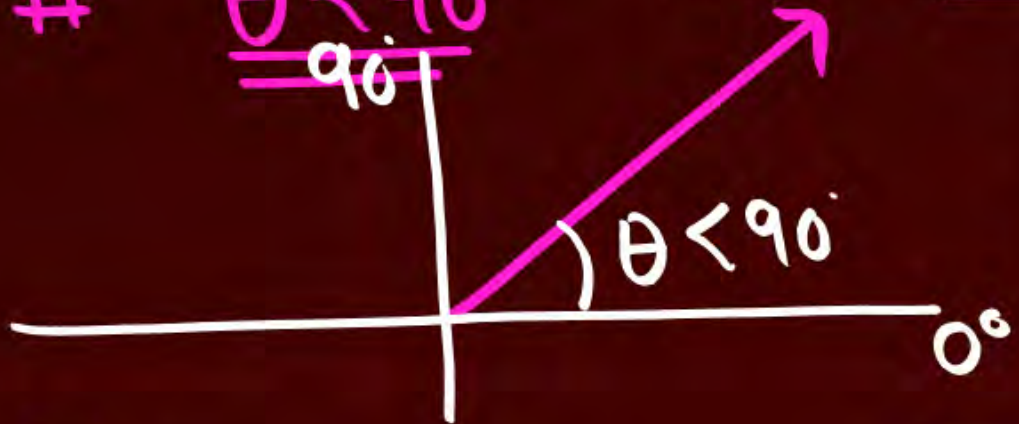
$$\text{Slope} = 2$$

Graph

Slope = +ive

$\theta < 90^\circ$
90°

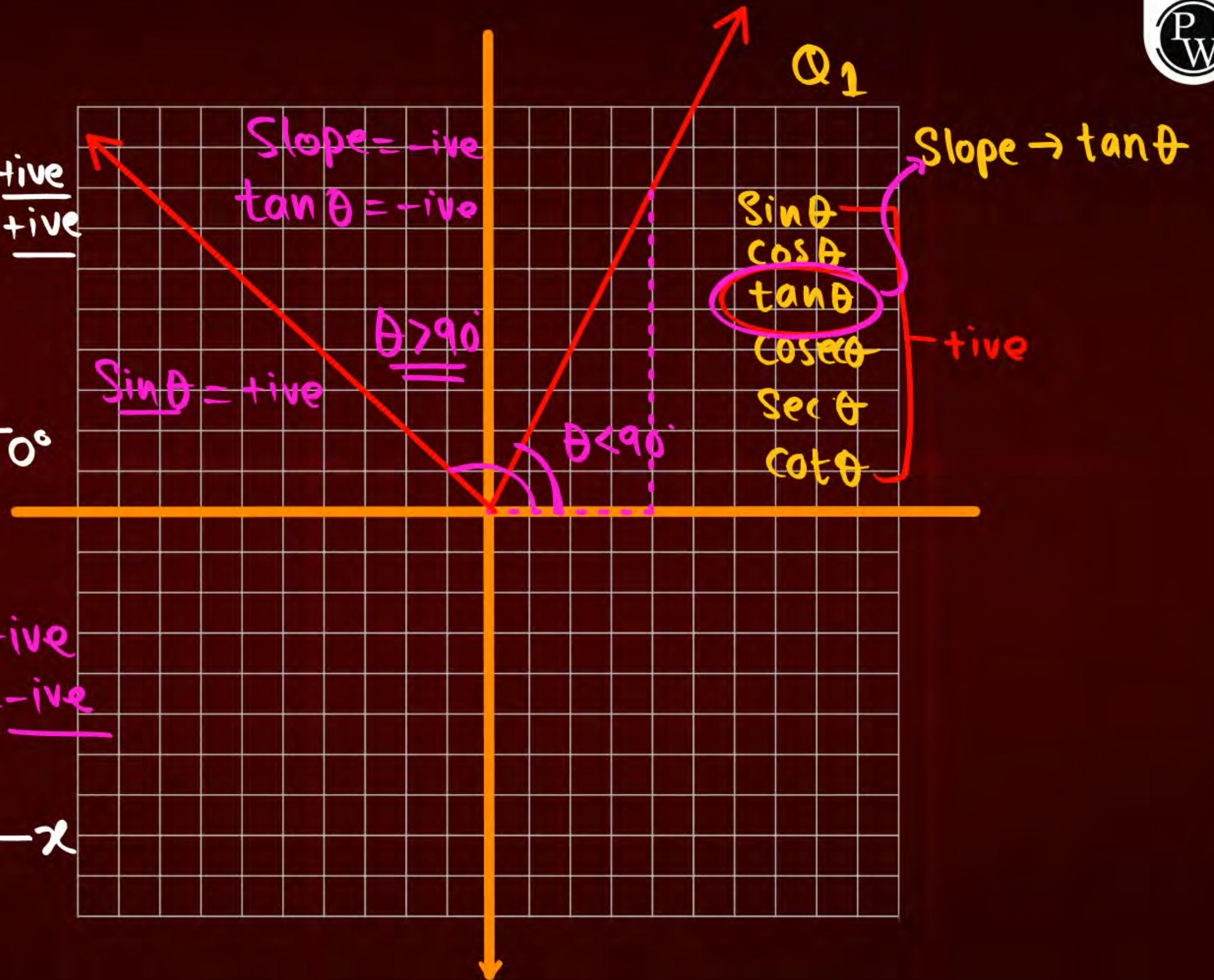
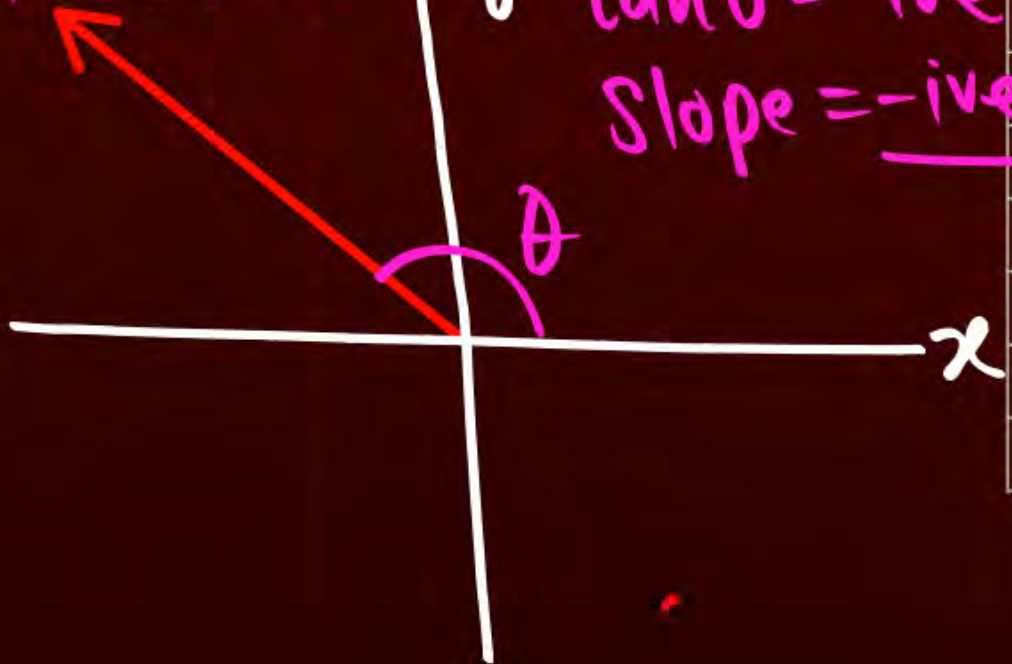
$\tan \theta = \underline{\text{five}}$
Slope = five

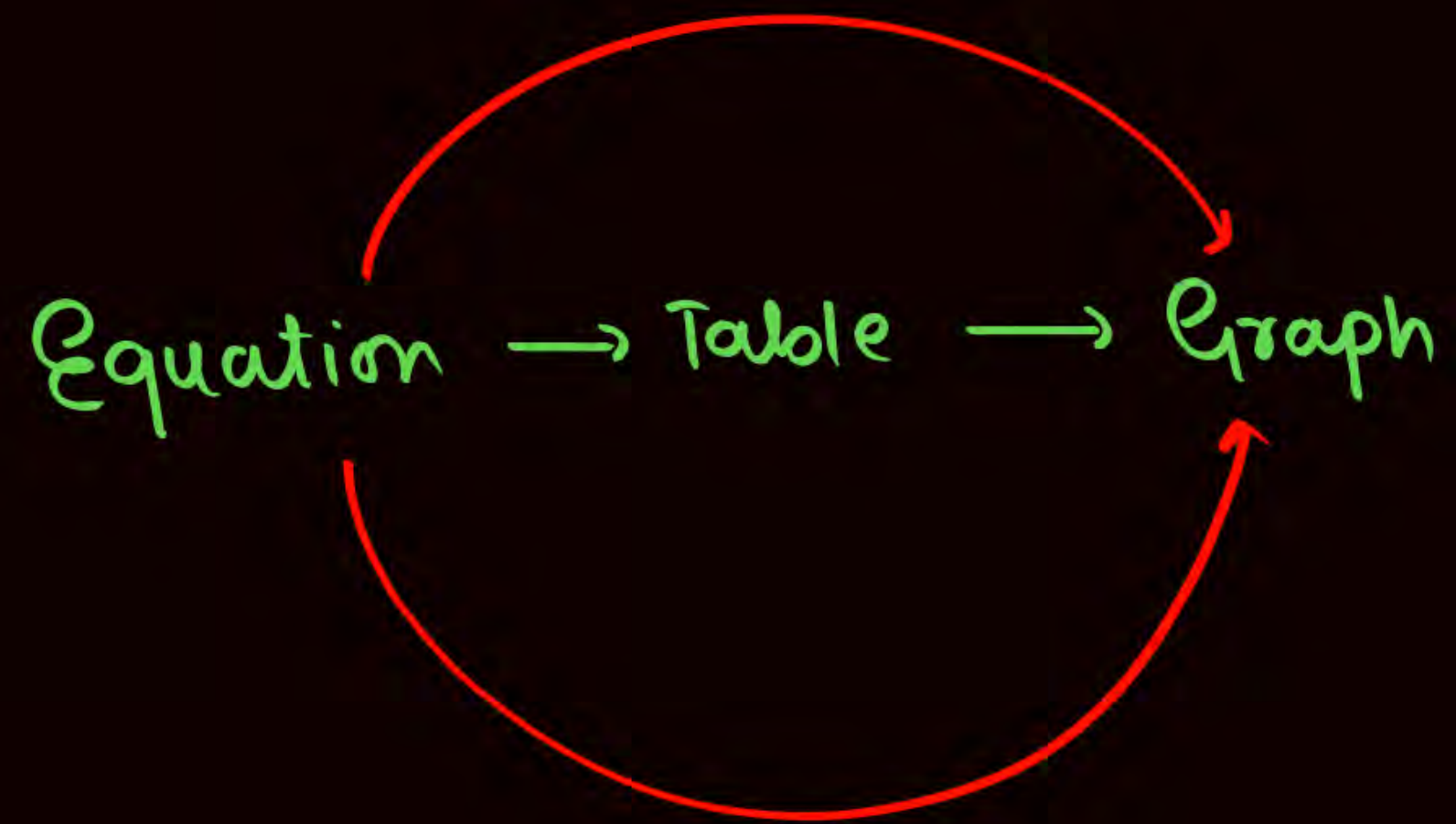


Slope = -ive

$\theta > 90^\circ \uparrow y$

$\tan \theta = -\text{ive}$
Slope = -ive

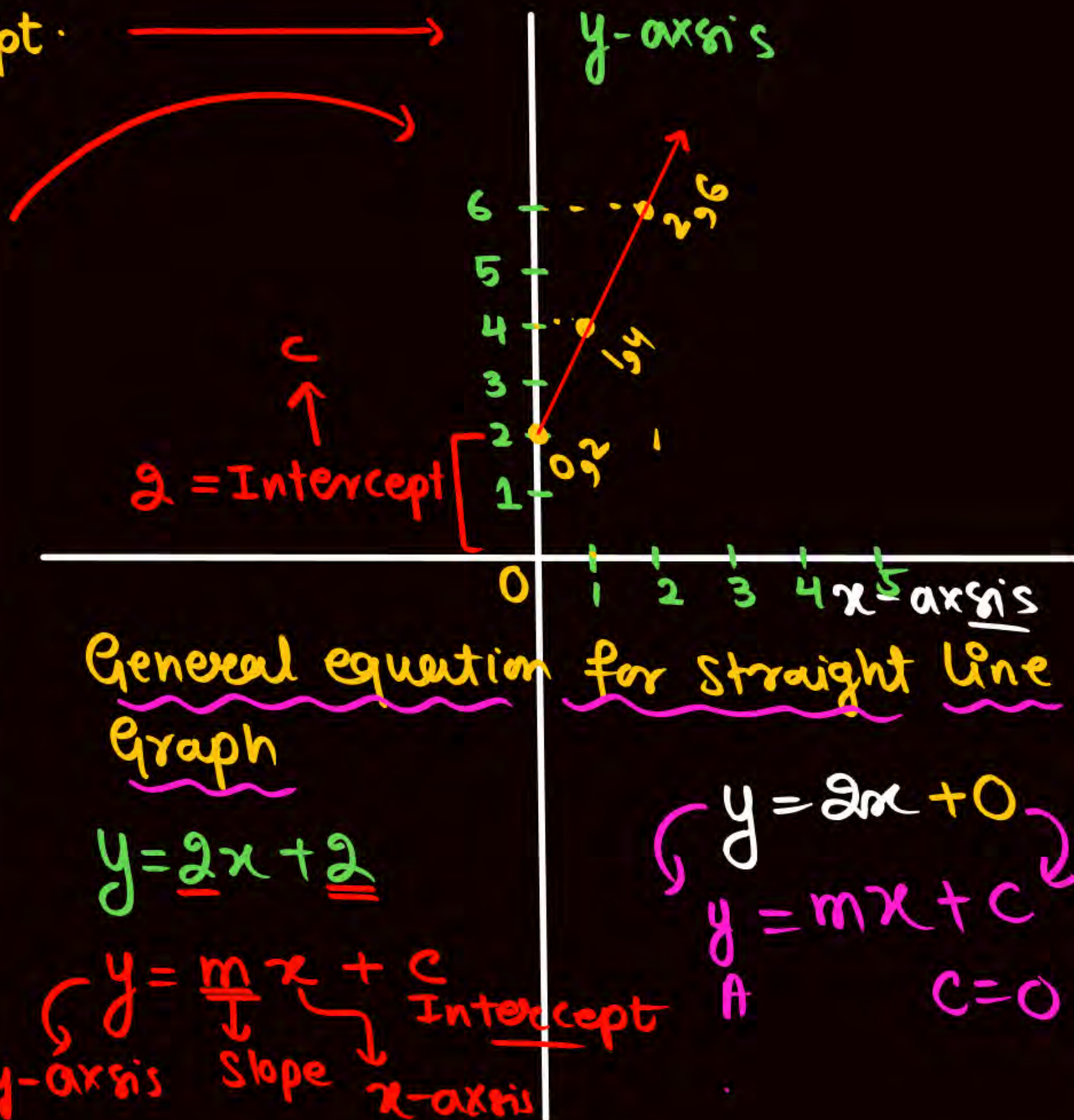




- $y = 2x + 2 \rightarrow$ y-intercept.
 $\checkmark x=0$

x	0	1	2
y	$y = 2x + 2$ $y = 2 \times 0 + 2$ $y = 0 + 2$ $y = 2$	$y = 2x + 2$ $y = 2 \times 1 + 2$ $y = 2 + 2$ $y = 4$	$y = 2x + 2$ $y = 2 \times 2 + 2$ $y = 4 + 2$ $= 6$

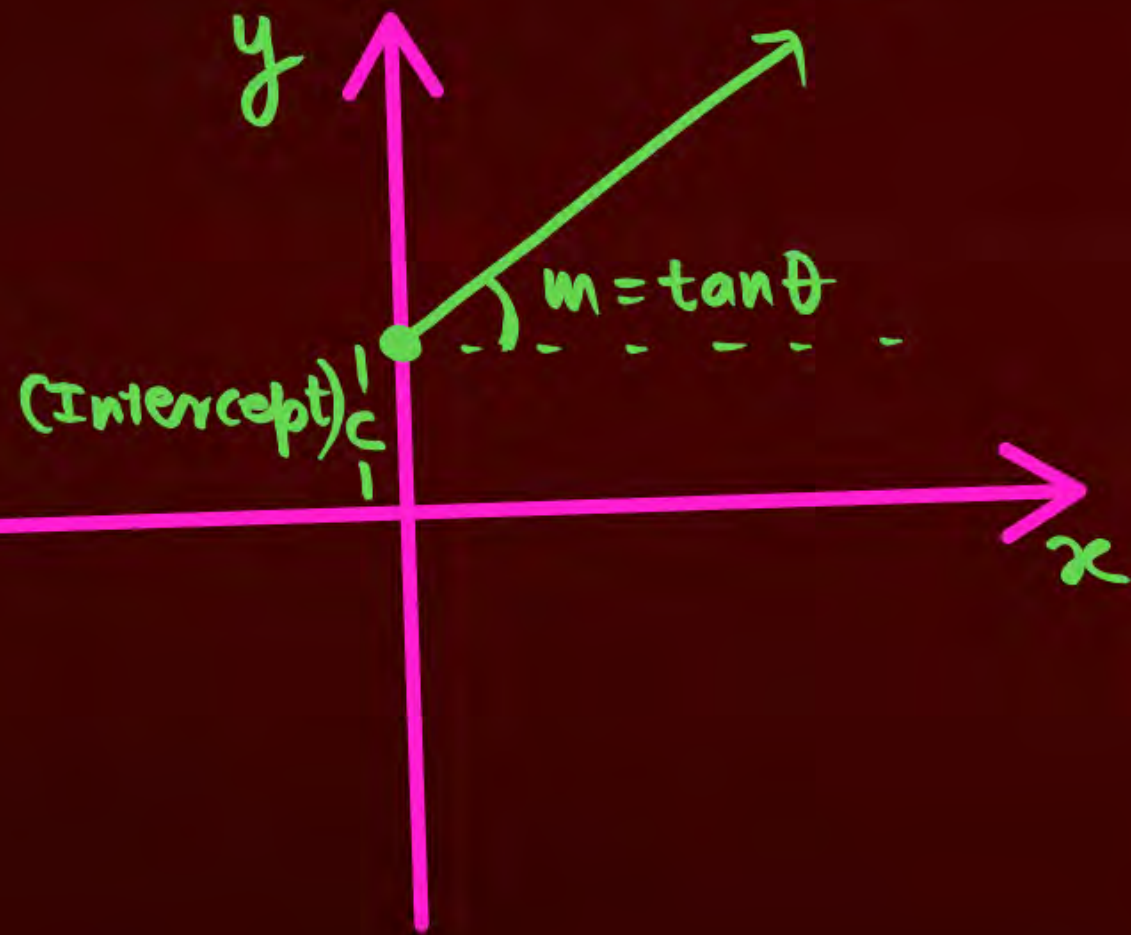
Intercept



Graph

1. Straight line graph

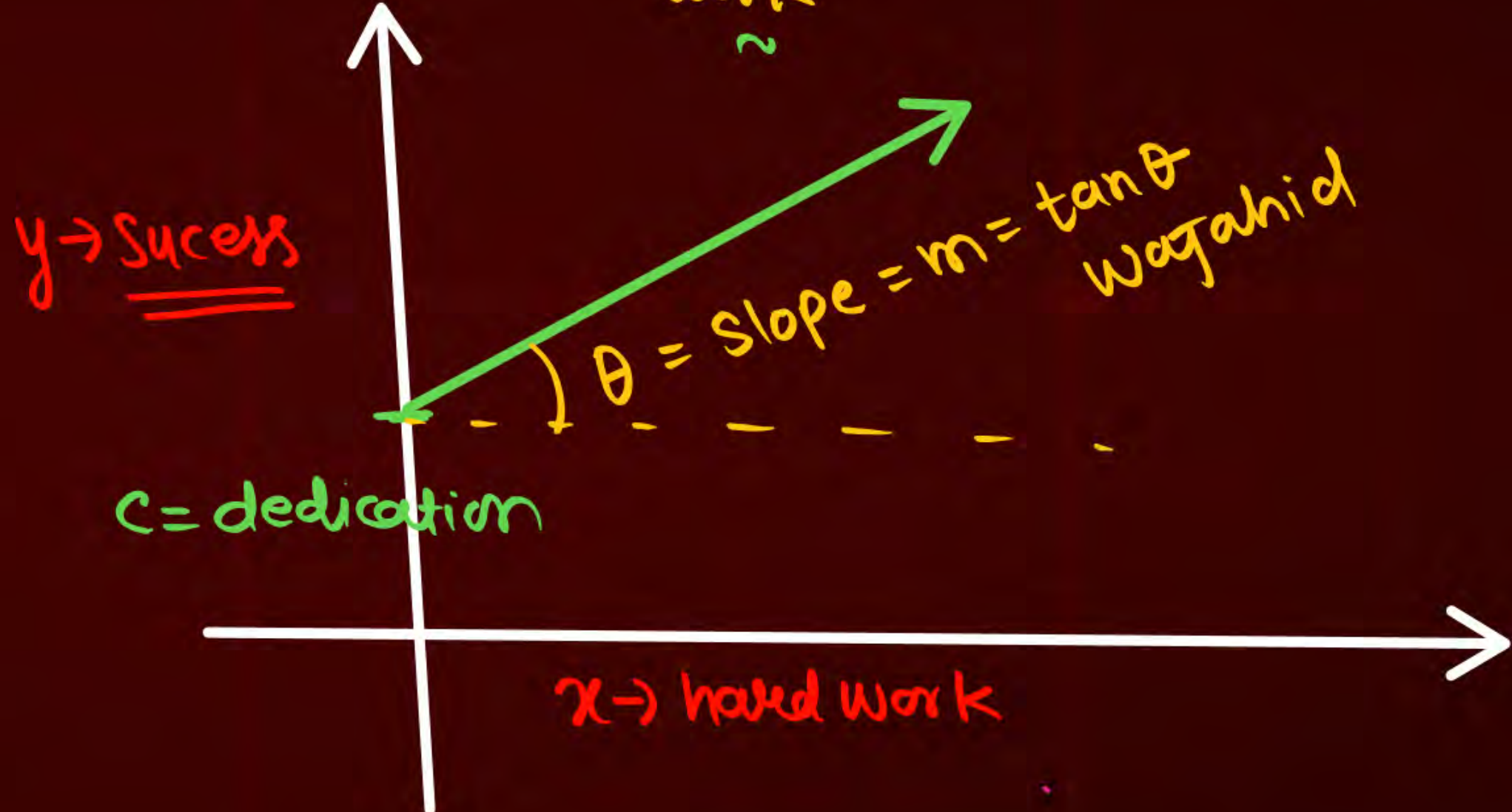
$$y = mx + \underline{c}$$



Slope = Wajahid Sir

$$\underline{y} = \underline{m}x + \underline{c}$$

Success
Hard work
dedication

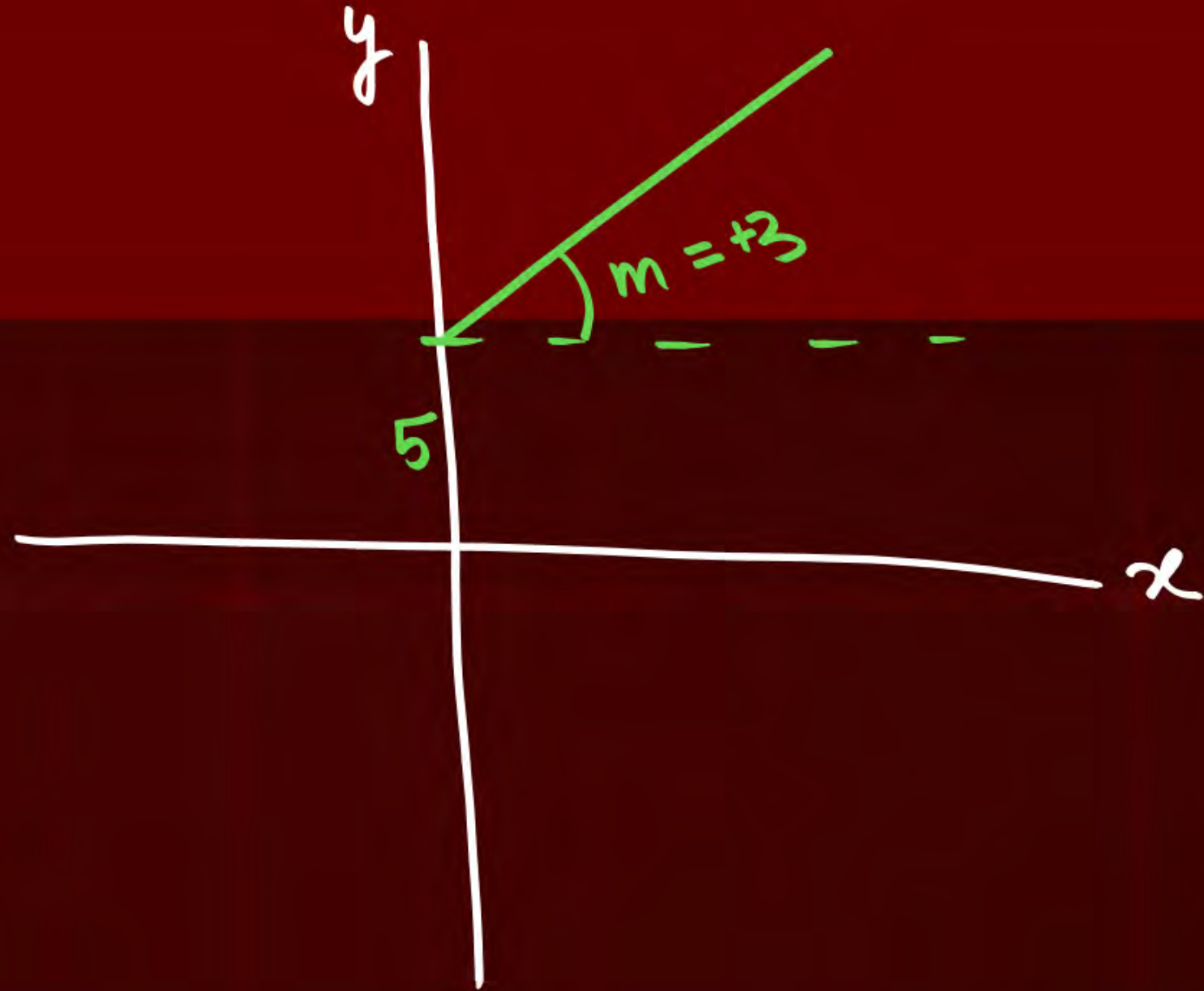


QUESTIONS

$$y = \underline{3}x + 5$$

$$\text{Slope} = 3 \checkmark$$

$$c = 5 \checkmark$$



QUESTIONS

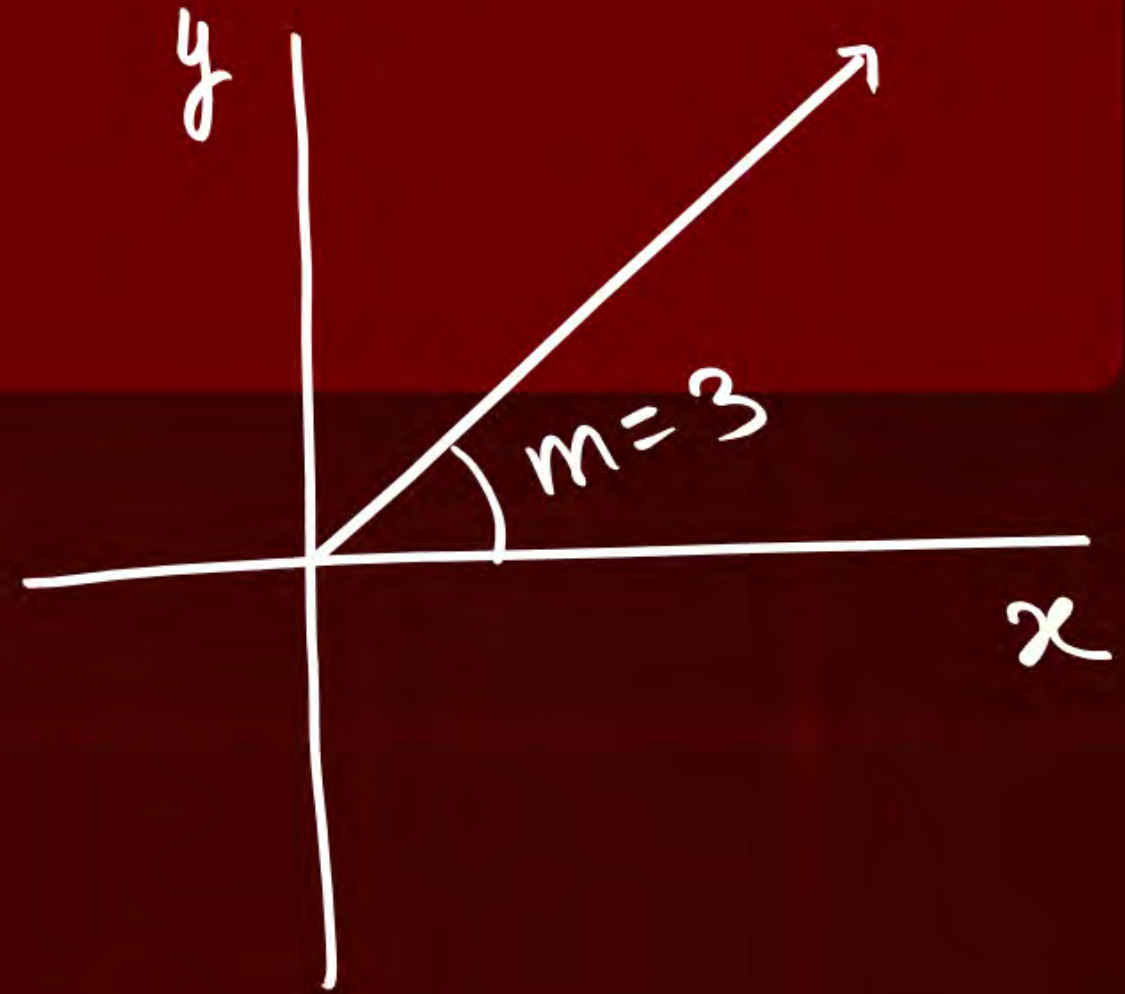
$$y = 3x, \quad C = 0$$

$$y = 3x + 0$$

$$\text{Slope} = 3$$

$$y = 3x + 0 \quad C = 0$$

$\uparrow \qquad \qquad \uparrow$
 $y = mx + c$



QUESTIONS

$$y = 3x - 2,$$

$$y = 3x - 2$$

| ↗

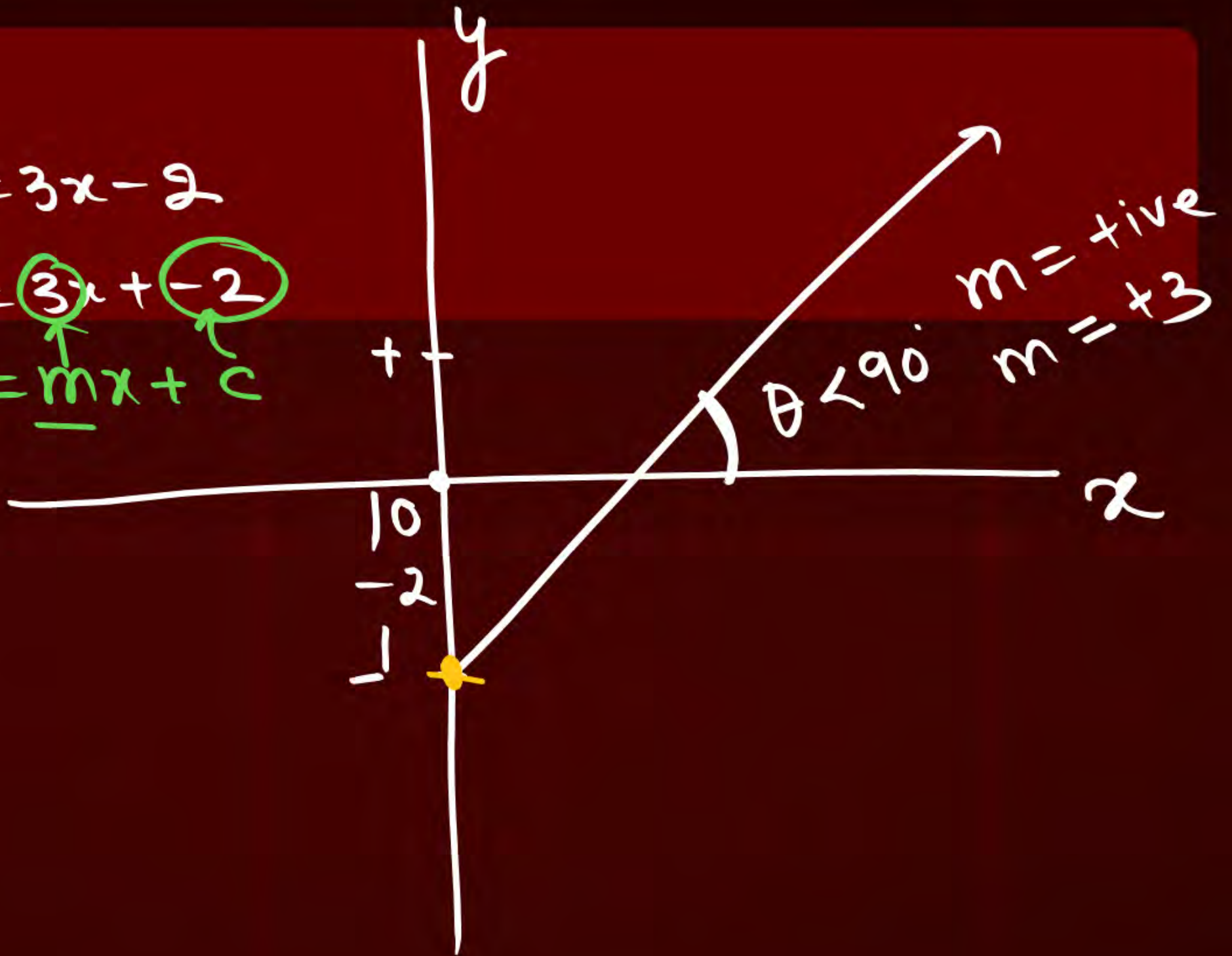
$$y = mx + c$$

$$y = 3x - 2$$

$$y = \textcircled{3}x + \textcircled{-2}$$

↑ ↗

$$y = \underline{m}x + c$$



QUESTIONS

$$y = \underline{-3}x + \underline{2}$$

$$\text{Slope} = -3$$
$$c = +2$$



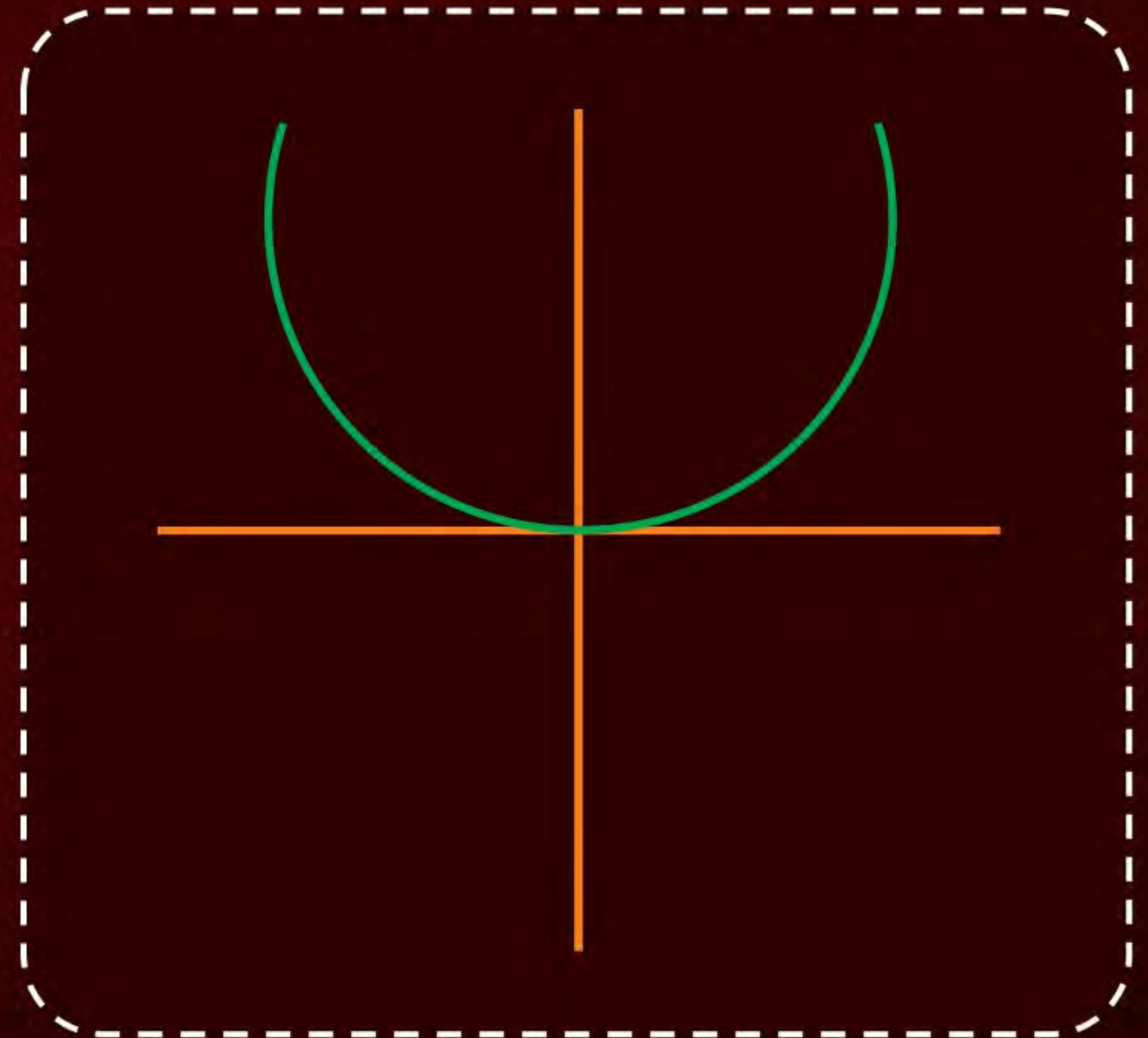
Graph

2. $y = x^2$

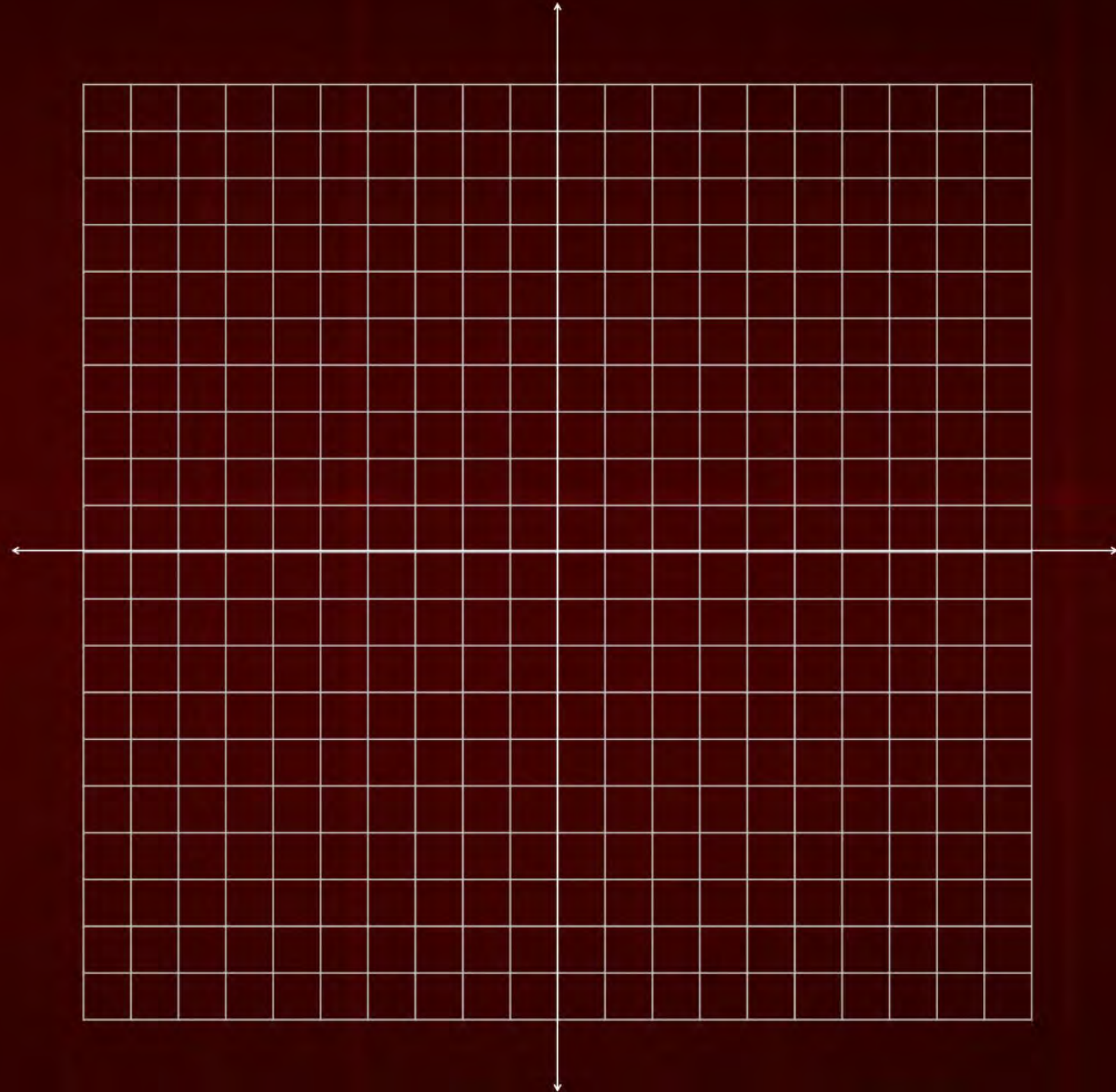
QUESTIONS

See Graph

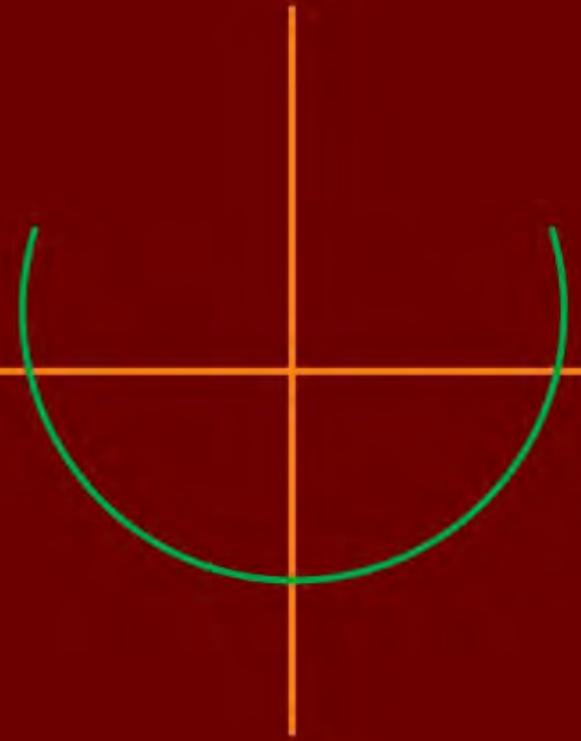
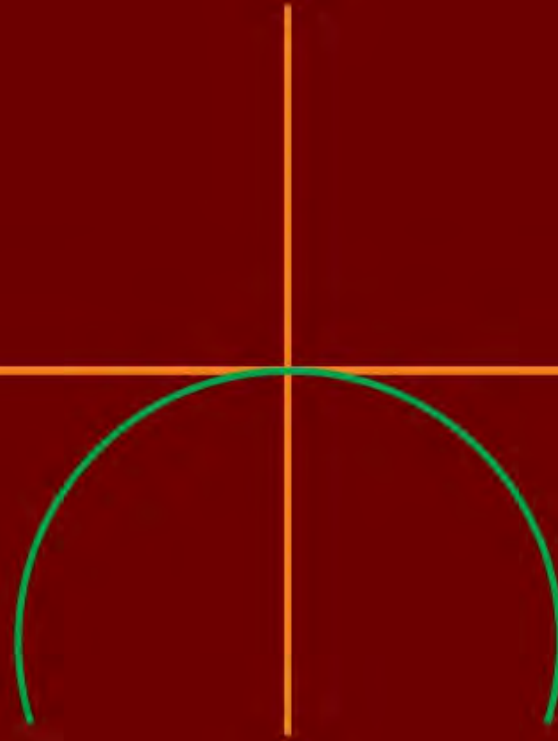
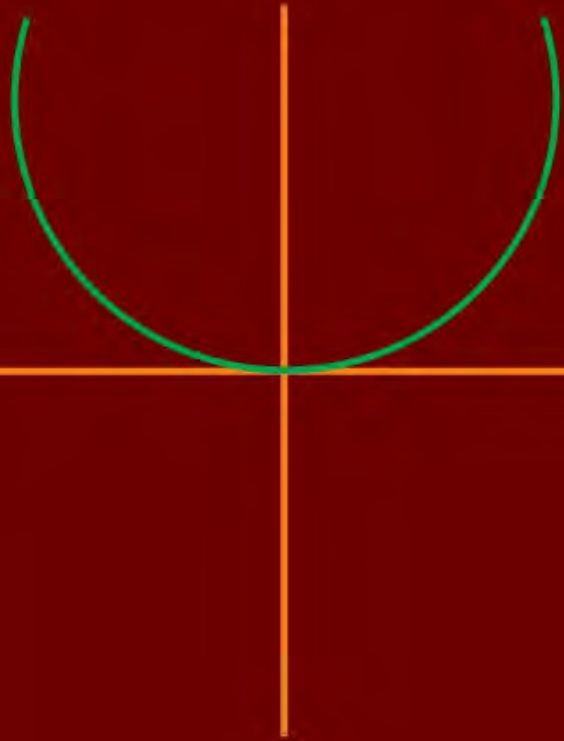
- 1 $y = x^2$
- 2 $y = mx + c$
- 3 $y = mx$
- 4 $y = \frac{1}{x}$



Graph



QUESTIONS



SUMMARY



Straight line graph

• Division

$$y = mx + c$$



Small Angle Approx.

Table

equation

Graph

QUESTIONS

$$y = -3x - 3,$$

HW

$$m = 9$$

$$c = ?$$

Thank
You